Product Requirements Document (PRD)

SmartLend — AI-Powered Lending Transparency Platform

Voice-First Agentic Architecture for Women Micro-Entrepreneurs

Version	3.0 (Agentic + Voice-First Pivot)
Status	Draft / Discovery Phase
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Target Audience	Engineering, Product, AI Ethics
Primary Persona	Mann Deshi Small Business Women

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1. Executive Summary

SmartLend is evolving from a standalone mobile audit app into a **voice-first, agentic AI system** designed to empower women micro-entrepreneurs in India's self-help group (SHG) and cooperative lending ecosystem. The platform audits loan agreements, detects predatory terms, validates NBFC legitimacy, provides daily bookkeeping support, and assists borrowers in understanding their rights under RBI's Digital Lending Guidelines 2022.

What Changed in Version 3.0

Persona Focus: Narrowed to a single MVP persona — the Mann Deshi small business woman (kirana shop owners, livestock managers, tailors) — based on Priyanka Kamath's field-validated recommendation across data availability, ease of implementation, and network effect dimensions.

Agentic Architecture

Replaced monolithic audit engine with a modular agent that dispatches to specialized skills (Loan Extractor, NBFC Verifier, Rate Calculator, Red Flag Scanner, Bookkeeping Assistant) via raw function calling. Model-agnostic design with Google Gemini as the recommended default for native multilingual support (Marathi, Hindi, Telugu).

Voice-First + Multi-Channel

Voice is the primary interface, not an add-on. Three channel options are presented — WhatsApp, IVR (voice call), and native app — with final channel selection pending team input. Bookkeeping moves into MVP as the daily-engagement hook.

Channel Selection :

Finalized Whatsapp, as there is no learning curve to ensure that it best serves the Mann Deshi demographic.

2. Customer Problem Statement

India's digital lending boom has disbursed over 11 crore loans as of 2026, but this growth has created a parallel crisis of borrower exploitation that disproportionately impacts women in semiurban and rural India.

2.1 The Systemic Problem

- **Predatory Lending:** Interest rates from unregulated lenders range from 18–26% or higher, with hidden processing fees, insurance charges, and penalty structures buried in fine print.
- **Fraudulent Apps:** Fake fintech firms have partnered with NBFCs using false addresses, leading to coercive recovery tactics and data misuse.
- **Financial Illiteracy Barrier:** Over 60% of microfinance borrowers lack understanding of loan terms, leading to over-indebtedness and debt traps.
- **Regulatory Opacity:** Borrowers cannot easily verify if a lending app is RBI-registered or compliant with Digital Lending Guidelines.

2.2 The Specific Problem for Mann Deshi Women

Within the broader crisis, women running small businesses in semi-urban Maharashtra and similar regions face compounded challenges:

- **Literacy Barrier:** Written loan agreements in English or formal Hindi are inaccessible. These women operate in Marathi and other vernacular languages, often with limited text literacy.
- **App Installation Barrier:** Many target users are reluctant to install new apps, lack device storage, or use low-end smartphones. Any solution must work on the platforms they already use.

- **No Structured Financial Records:** Kirana shop owners, tailors, and livestock managers make financial decisions daily (pricing, credit to customers, loan repayments) but lack tools to track cash flow — making them vulnerable to over-borrowing.
- **SHG Group Dynamics:** Financial decisions often happen at the group level. One SHG leader may handle loan applications for 10–15 members, yet existing tools are designed for individual use.

3. Why We Are Building This

- **Financial Safety:** Protecting borrowers from predatory terms that could cost thousands of rupees in hidden fees and excessive interest.
- **Regulatory Empowerment:** Making RBI guidelines actionable by translating complex regulatory language into voice-delivered, vernacular alerts.
- **Daily Utility via Bookkeeping:** Loan auditing is episodic (1–2x/year), but bookkeeping is daily. The bookkeeping skill creates habitual engagement and builds the trust relationship that makes users turn to the agent when they need a loan audited.
- **SHG Network Effects:** RBI and SHG ecosystem data shows one financially empowered woman typically influences 3–5 others in her community. Mann Deshi's existing SHG network makes this persona a force multiplier.
- **Voice-First Inclusion:** Removing the literacy barrier entirely. The system speaks Marathi, Hindi, and Telugu natively — voice in, voice out.
- **Zero-Friction Access:** Meeting users where they already are — whether that's WhatsApp, a phone call, or an Android app — without requiring new app installation or complex onboarding.

4. User Persona

4.1 MVP Persona: The Mann Deshi Small Business Woman

Dimension	Detail
Name	Meena (representative persona)
Demographics	Woman, age 25–45, semi-urban/rural Maharashtra or similar region
Occupation	Kirana (grocery) shop owner, tailor, livestock manager, or similar microenterprise
Financial Context	Engages with SHG/cooperative lending; manages daily bookkeeping, pricing, and credit decisions; takes microfinance loans for inventory or agricultural inputs
Banking	Jan Dhan account holder (high penetration, low active usage); may use Mann Deshi Mahila Bank or similar women's cooperative
Technology	Smartphone (budget Android); fluent in WhatsApp voice notes; limited comfort with text-heavy apps; prefers Marathi or Hindi

Key Need	Vernacular voice interface to audit loan terms, verify lender legitimacy, compare rates against fair benchmarks, and track daily income/expenses
Why This Persona?	Richest behavioral data (repeat loans, savings, inventory, repayments); AI integration maps to existing daily routines; highest network-effect multiplier via SHG ecosystem
Data Source	Priyanka Kamath's field assessment + Mann Deshi Mahila Bank program data + RBI financial inclusion data + Mastercard/CARE Strive Women Baseline Report (May 2025)

4.2 Future Personas (Post-MVP)

The following personas from PRD v1.0 remain valid but are explicitly deferred to post-MVP phases:

- **Priya (Young Professional):** First-time borrower seeking personal loan via digital NBFC. Needs plain-language APR explanation. Channel: native app.
- **Rajesh (Small Business Owner):** Taking business loan from fintech for inventory. Needs lender legitimacy verification. Channel: WhatsApp.
- **Arjun (Urban Gig Worker):** Considering BNPL/quick loan for emergency. Needs immediate red-flag detection. Channel: WhatsApp or app.
- **Sita (College Student):** Learning personal finance and government schemes. Needs financial literacy education. Channel: WhatsApp.

5. System Architecture: Agentic Design

SmartLend v3 replaces the monolithic audit engine with a **conversational agent backed by modular, independently deployable skills**. The agent handles intent recognition and conversation flow, then dispatches to specialized skills via raw function calling.

5.1 Design Principles

- **Model-Agnostic Core:** The orchestrator and skills are designed to work with any LLM that supports function calling. Google Gemini is the recommended default for its native multilingual capability (Marathi, Hindi, Telugu, Bengali) and competitive pricing. Alternative models (Claude, GPT-4o, open-source) can be swapped without architectural changes.
- **Raw Function Calling for MVP:** No framework overhead. The agent uses the LLM's native function-calling API to dispatch to skills. This keeps the system simple, debuggable, and fast. Framework adoption (LangGraph, etc.) deferred to post-MVP if multi-step workflow complexity demands it.
- **Skill Independence:** Each skill is independently testable, deployable, and improvable. New skills can be added without redesigning the system.
- **Voice-First Pipeline:** Voice is the primary I/O modality, not an add-on. The architecture is designed around the flow: voice input → STT → intent + skill dispatch → response generation → TTS → voice output.

5.2 Agent Orchestrator

The orchestrator is the conversational brain that manages user sessions, maintains context, classifies intent, and dispatches to the appropriate skill via function calls.

- **Session Management:** Maintains per-user conversation state across turns, including language preference, ongoing audit context, and bookkeeping history.
- **Intent Classification:** Uses the LLM's native understanding to route user requests to skills. No separate intent classifier needed for MVP.
- **Multi-Skill Orchestration:** A single user request may trigger multiple skills in sequence (e.g., "check this loan" triggers Loan Extractor → NBFC Verifier → Rate Calculator → Red Flag Scanner → Advisory response). Raw function calling handles this as sequential tool calls within a single LLM turn.
- **Language Handling:** Gemini processes Marathi, Hindi, and Telugu natively. No translate-first approach needed — the LLM reasons directly in the user's language.

5.3 Skills (Modular Tool Belt)

Each skill is a self-contained function the agent can call. MVP includes five skills:

Skill	Input	Processing	Output
Loan Term Extractor	Photo/PDF of loan agreement sent as image	Vision model (Gemini Flash) performs OCR + structured extraction in Hindi/Marathi/English	JSON with lender name, NBFC reg #, principal, interest rate, tenure, EMI, fees, penalties

NBFC Verifier	Lender name and/or registration number from extractor	Lookup against cached RBI registered NBFC database (updated monthly)	Legitimacy status: Registered / Not Found / Regulatory Action flagged
Rate Calculator	Interest rate (flat or APR), principal, tenure, fees from extractor	Deterministic calculation: flat-to-APR conversion, all-in-cost computation, benchmark comparison	True APR, total cost breakdown, comparison against RBI repo rate + fair margin, market rate comparison
Red Flag Scanner	Extracted terms and clauses from loan agreement	Pattern matching against known predatory clause templates + LLM analysis for context	List of red/yellow/green flags with plain-language explanation for each
Bookkeeping Assistant	Voice note: “aaj 500 rupaye ki sabzi bechi” (sold 500 rupees of vegetables today)	STT → LLM extracts amount, category, income/expense → appends to user’s ledger	Running daily/weekly/monthly summary; cash flow visibility; savings progress against goals

Post-MVP skills to consider: Grievance Filing Assistant (draft RBI Ombudsman complaint), Government Scheme Matcher (surface relevant central/state schemes), Credit Health Monitor (via Credit Bureau API with consent), Loan Comparison Engine (alternative lender suggestions).

5.4 Voice Pipeline (Gemini End-to-End)

The voice pipeline is designed for Gemini’s native multimodal and multilingual capabilities, enabling a single-model flow from speech to reasoning to speech.

- **Step 1 — Speech-to-Text:** User sends voice input (WhatsApp voice note, IVR audio, or in-app recording). Gemini’s native audio understanding transcribes Marathi, Hindi, Telugu, and Hinglish directly — no separate STT service needed for MVP.
- **Step 2 — Reasoning + Skill Dispatch:** The transcribed text (or raw audio, if using Gemini’s audio-native mode) is processed by the orchestrator. Intent is classified, skills are dispatched via function calling, and results are synthesized into a vernacular response.
- **Step 3 — Text-to-Speech:** Response is converted to audio in the user’s language. For WhatsApp: returned as a voice note. For IVR: streamed as real-time audio. For native app: played through device speakers.
- **Fallback STT:** If Gemini audio input is unavailable or for cost optimization, Whisper or Google Speech API can serve as fallback STT with the transcription routed to Gemini for reasoning.

Why Gemini End-to-End?

Gemini’s native Marathi, Hindi, and Telugu comprehension eliminates the translate-first antipattern (vernacular → English → reason → English → vernacular) that degrades accuracy and adds latency. A single model that reasons natively in the user’s language

produces more natural, contextually appropriate responses — critical for building trust with first-time users.

5.5 Financial Literacy Layer (DhanKanya Integration)

The DhanKanya open-source project (github.com/100-GIGA/DhanKanya) provides the reference architecture for SmartLend's financial education layer. Key components adopted: •

- **RAG-Powered Q&A:** ChromaDB vector store with HuggingFace BGE embeddings (BAAI/bge-small-en-v1.5). Knowledge base includes RBI Digital Lending Guidelines 2022, Fair Practices Code for NBFCs, predatory clause patterns, government financial schemes for women, and basic financial literacy corpus.
- **State-Specific Guidance:** Adapted from DhanKanya's state_templates.json — regional NBFC complaint mechanisms, local interest rate benchmarks, state government schemes, and consumer protection resources.
- **Fallback Behavior:** When vector similarity score falls below 0.7, queries route directly to the LLM with lending-domain system prompts.

6. Interface Channels

FINAL CHANNEL STRATEGY: WHATSAPP BUSINESS API

The final channel strategy inputs are in tandem with Priyanka Kamath and 100 GIGA team, who have direct field knowledge of the Mann Deshi demographic's technology usage patterns. This section presents three options with their tradeoffs for the team to evaluate.

6.1 WhatsApp Business API

Strengths

- 500M+ Indian users; zero installation barrier; users already send voice notes daily
- WhatsApp's native voice messages map directly to voice-first design
- SHG group chats provide natural distribution channel
- Rich media support: receive loan document photos, send PDF reports, use interactive list messages

Constraints and Risks

- **24-Hour Session Window:** WhatsApp Business API only allows free-form messages within 24 hours of the user's last message. After that, only pre-approved template messages can be sent. This limits proactive follow-ups (e.g., "your monthly cash flow summary is ready").
- **Template Message Approval:** Every proactive message template must be submitted to Meta for approval. Hindi/Marathi templates have longer approval cycles. Templates cannot contain dynamic financial advice.
- **Media Size Limits:** Incoming images capped at 5MB; documents at 100MB. Low-quality phone camera photos of loan agreements may exceed 5MB. Compression or re-capture guidance needed.
- **Privacy Tradeoff:** Documents transit through Meta's infrastructure and SmartLend's backend — a departure from the original on-device privacy promise. Requires explicit user consent, encryption at rest (AES-256), and auto-deletion within 24 hours.
- **Meta Policy Risk:** Meta's updated WhatsApp Business API terms (January 2026) prohibit general-purpose AI chatbot providers. SmartLend is a business serving customers (not an AI provider), but policy interpretation could shift. Requires ongoing compliance monitoring.
- **Per-Message Cost:** BSP fees (Gupshup, Twilio, WATI) add per-message cost. For a non-profit/low-margin model, this needs careful budgeting. Estimated: ₹0.50–1.50 per conversation.
- **SHG Group Mode Complexity:** A group leader auditing loans on behalf of 10–15 SHG members needs multi-user state management under a single WhatsApp number. Architecturally doable but must be designed from the start.

6.2 Recommended Phasing

Based on the Mann Deshi persona's characteristics, the team has locked in on the validation:

- **MVP (Months 1–2):** WhatsApp handles document-based loan audits and bookkeeping.
- **Phase 2 (Months 3–4):** Evaluate channel usage data. Double down on the channel showing higher engagement and completion rates.
- **Phase 3 (Months 5+):** Native Android app for users who want offline access, visual dashboards, and premium features. Positioned as the "upgrade" path.

Update :

The end user we are considering here is a rural village women using services of Mann Deshi Mahila Sahakari Bank which is pioneering a **chat-first banking interface** for underserved women entrepreneurs in India by leveraging **WhatsApp** as the primary distribution layer. The model replaces:

- Branch-heavy onboarding
- App-based UX barriers
- Document-heavy underwriting
- With **AI-assisted, conversational financial services delivered inside a messaging platform.**

We as Team B2 + 100 GIGA need to explore what additionally are we bringing to the table.

7. Functional Specifications

7.1 Loan Document Audit Flow

The primary use case. User sends a loan agreement; the agent returns a plain-language risk assessment in the user's vernacular.

- **Input Formats:** Photos of physical documents, PDF loan agreements, screenshots of app-based loan offers (WhatsApp/app); voice-described loan terms (IVR)
- **Extraction Fields:** Lender name and NBFC registration number, principal amount, interest rate (flat and APR), tenure and EMI, processing fees/GST/insurance, penalty clauses, sanction date and disbursement details
- **Language Support:** Marathi, Hindi, Telugu, English, and Hinglish (code-mixed) via Gemini's native multilingual vision and text capabilities
- **Accuracy Target:** 95% field-level extraction accuracy on standard NBFC loan agreement templates

7.2 Audit Engine (Orchestrated via Skills)

The audit engine is not a monolith — it is the orchestrator dispatching to the NBFC Verifier, Rate Calculator, and Red Flag Scanner skills in sequence.

A. Lender Verification (NBFC Verifier Skill)

- Cross-reference lender against RBI's registered NBFC database (cached, updated monthly)
- Flag unregistered lenders or those with regulatory actions/warnings
- Verify LSP (Lending Service Provider) registration if applicable

B. Interest Rate Analysis (Rate Calculator Skill)

- Convert flat interest rates to APR/reducing balance equivalent
- Compare against RBI repo rate + reasonable margin benchmarks
- Flag rates exceeding 26% APR as "High Risk — Potentially Predatory"
- Display comparison with average market rates for similar loan products

C. Hidden Charges Detection (Rate Calculator Skill)

- Identify processing fees exceeding 2% of principal
- Detect bundled insurance products not clearly disclosed
- Flag prepayment penalties exceeding RBI-permitted limits
- Calculate true "All-In Cost" including all fees and charges

D. Terms and Conditions Red Flags (Red Flag Scanner Skill)

- Scan for clauses requiring excessive app permissions (contacts, SMS, location)
- Identify aggressive recovery language or personal guarantor requirements
- Detect auto-debit without explicit consent mechanisms
- Flag loans requiring upfront payment before disbursement (classic fraud indicator)

7.3 Advisory Response

- **Risk Score:** Low/Medium/High with clear vernacular reasoning
- **Vernacular Explanations:** Complex financial terms translated and voice-delivered in user's language
- **Action Recommendations:** "Negotiate processing fee reduction," "Request APR-based interest," "Avoid this lender — not RBI registered," "Consider bank personal loan for lower APR"
- **Grievance Guidance:** If fraudulent lender detected, provide RBI Ombudsman filing instructions and draft complaint (via future Grievance Filing skill)

7.4 Bookkeeping Assistant (MVP Skill)

The bookkeeping skill is the **daily-engagement hook** that builds habitual usage between episodic loan audits. It maps directly to the Mann Deshi persona's existing daily routines.

- **Voice-First Logging:** User sends a voice note: "aaj 800 rupaye ka doodh becha" (sold 800 rupees of milk today). Agent extracts amount, category (income/expense), and appends to the user's ledger.
- **Daily/Weekly/Monthly Summaries:** On-demand or proactive (via template message) cash flow summaries in vernacular.
- **Savings Goals:** Set targets ("I want to save 5000 rupees this month for inventory") with progress tracking.
- **Post-Audit Integration:** After a loan audit calculates the true EMI and all-in cost, the bookkeeping module pre-populates the EMI as a recurring expense for immediate affordability assessment.
- **SHG Group Bookkeeping:** The SHG group leader can log group-level transactions (collections, disbursements) and generate group financial summaries. This is architecturally significant — see Section 6.1 on multi-user state management.

8. Technical Performance Targets

Metric	Target	Notes
Voice Response Latency	< 3 seconds	STT + LLM reasoning + TTS; critical for natural conversation feel
Loan Document Extraction	< 5 seconds	Vision model OCR + structured extraction
Full Audit (end-to-end)	< 30 seconds	Extraction + all skill dispatches + advisory generation
Bookkeeping Entry	< 2 seconds	Voice note → parsed ledger entry
RAG Retrieval Latency	< 500ms	ChromaDB vector similarity search
Interest Rate Calculation	99% accuracy	Deterministic; no LLM involved

Lender Verification	95% match rate	Against RBI database
Red Flag Detection	90% precision	Pattern + LLM analysis
Vernacular Comprehension	90%+ accuracy	Marathi, Hindi, Telugu voice input understanding
WhatsApp Acknowledgment	< 3 seconds	First response after user message
IVR Connection	< 5 seconds	Ring-to-agent-ready latency

9. Success Metrics

9.1 User Protection Metrics

- **Fraud Detection Rate:** Percentage of unregistered/fraudulent lenders successfully flagged in test dataset
- **Cost Savings Identified:** Average rupee value of hidden charges detected per loan audit
- **User Empowerment Score:** Post-audit survey measuring user confidence in understanding loan terms (target: 80% report feeling “more confident”)

9.2 Engagement Metrics

- **Daily Active Bookkeeping Users:** Percentage of users who log at least one income/expense entry per day (target: 40% of registered users)
- **Audit-to-Bookkeeping Conversion:** Percentage of users who start with a loan audit and adopt daily bookkeeping within 14 days (target: 25%)
- **Voice vs. Text Ratio:** Percentage of interactions via voice (target: >70% for Mann Deshi persona, validating voice-first design)

9.3 SHG Network Metrics

- **SHG Group Adoption:** Number of SHG groups where at least one member is an active user
- **Referral Multiplier:** Average number of new users acquired per existing SHG user (target: 3x, aligned with Priyanka’s 3–5 influence range)
- **Group Leader Usage:** Percentage of group leaders using the SHG group bookkeeping feature

9.4 Channel Metrics

- **Channel Adoption Split:** WhatsApp vs. IVR vs. app usage distribution across the Mann Deshi demographic
- **Audit Completion Rate:** Percentage of document submissions resulting in complete audit report (target: 85% on WhatsApp, 70% on IVR)
- **Conversation Depth:** Average follow-up messages per audit session (target: 3+)

9.5 Privacy and Compliance

- **WhatsApp Auto-Deletion Compliance:** Verified via automated testing that no PII persists beyond 24-hour window
- **User Consent Rate:** Percentage of first-time users who complete the privacy disclosure and opt-in flow

10. Ethical and Legal Guardrails

10.1 Regulatory Compliance

- **RBI Alignment:** All audit criteria align with RBI's Digital Lending Guidelines (2022) and Fair Practices Code
- **DPDPA 2023:** Compliance via ephemeral cloud processing with auto-deletion (WhatsApp/IVR) or on-device processing (native app)
- **WhatsApp Business API Compliance:** SmartLend positioned as a business serving customers (not a general-purpose AI provider). Regular review of Meta's evolving API policies required.
- **No Unauthorized Financial Advice:** Persistent disclaimer in all channels: "This is an educational tool providing information based on RBI guidelines. It does not constitute financial or legal advice."

10.2 AI Ethics Safeguards

- **Hallucination Prevention:** If loan term cannot be interpreted with confidence, system displays "Unable to analyze — manual review recommended" rather than generating false analysis
- **Bias Mitigation:** Audit logic does not discriminate based on loan amount, borrower geography, or lender size
- **Explainability:** Every risk flag includes clear explanation: "This charge is flagged because it exceeds RBI-permitted limit of X%"
- **Human-in-the-Loop:** Agent cannot auto-submit complaints or communicate with lenders on user's behalf; all actions require explicit user confirmation

10.3 Accessibility and Inclusion

- **Voice-First as Default:** Not an accessibility feature — the primary interface. Removes literacy barrier entirely.
- **Vernacular Priority:** Marathi, Hindi, Telugu as launch languages; 5+ Indian languages in roadmap
- **Low-Literacy Design:** Visual risk indicators (red/yellow/green), voice-delivered explanations, minimal text
- **Offline-First (Native App):** Core features accessible without internet for rural users with limited connectivity

10.4 Lender Fairness

- **Evidence-Based Flagging:** Lenders flagged only on objective regulatory noncompliance or statistical anomalies
- **Monthly Updates:** NBFC registry and regulatory database refreshed monthly
- **Appeal Process:** Mechanism for lenders to report false flags via web portal

11. Go-to-Market Strategy

11.1 Target Launch Markets

- **Phase 1 (Months 1–3):** Mann Deshi Mahila Bank network in Maharashtra. Direct partnership with SHGs and cooperative lending groups. WhatsApp number and IVR tollfree number shared through existing SHG communication channels.
- **Phase 2 (Months 4–6):** Expand to other women's cooperatives and SHG networks in Maharashtra, then to neighboring states (Karnataka, Andhra Pradesh) with Telugu and Kannada language support.
- **Phase 3 (Months 7–12):** National scale via partnerships with NABARD, RBI Financial Literacy Centers, and microfinance networks. Tier 1 city expansion for additional personas.

11.2 Distribution Channels

- **SHG Network Distribution (Primary):** SmartLend's WhatsApp number and IVR number shared through SHG WhatsApp groups, printed on loan passbooks, and announced at SHG meetings
- **Mann Deshi Partnership:** Co-branded launch with Mann Deshi Mahila Bank; integration into their existing financial literacy programs
- **NGO Partnerships:** Collaboration with financial literacy NGOs, women's empowerment organizations, and microfinance networks
- **WhatsApp Viral:** WhatsApp's forward/share mechanism enables organic growth within SHG networks

11.3 Business Model (Post-MVP)

- **Freemium Core:** Loan audit and basic bookkeeping free forever across all channels
- **Premium Features:** Credit score monitoring, loan recommendations, priority support (₹99/month or ₹999/year) via native app
- **B2B Licensing:** White-label version for ethical NBFCs, cooperatives, and banks
- **Impact Partnerships:** Grants from financial inclusion foundations, CSR funding from banks, development finance institution funding

12. Risks and Mitigation

Risk	Impact	Mitigation

Meta WhatsApp API policy changes disable channel	**High**	Maintain IVR as independent parallel channel; native app as fallback; monitor Meta policy quarterly
IVR telephony costs exceed budget at scale	**Medium**	Negotiate volume discounts with India-focused providers (Exotel, Knowlarity); explore TRAI subsidized toll-free options for social impact
Gemini multilingual accuracy degrades for specific Marathi dialects	**High**	Model-agnostic architecture allows fallback to alternative models; collect dialect-specific evaluation data from Mann Deshi users; fine-tune if needed
SHG group leader becomes single point of failure	**Medium**	Design multi-admin capability; train 2–3 members per group; direct individual access always available
Voice pipeline latency exceeds conversational threshold	**High**	Pre-compute common responses; cache frequent NBFC lookups; optimize TTS streaming; set hard 3-second SLA with fallback to text
Predatory lenders pressure users to stop using service	**High**	In-app education on user rights; partnership with consumer protection orgs; anonymous usage mode
OCR accuracy fails on low-quality phone camera photos	**Medium**	Real-time image quality feedback; guide users to retake; voice-input fallback for loan terms
RBI regulatory changes invalidate audit logic	**Medium**	Monthly regulatory monitoring; push notifications for critical updates; version deprecation policy
Bookkeeping data loss or corruption	**High**	Encrypted local + cloud backup with user consent; daily ledger snapshots; conflict resolution for multi-device access

13. Development Roadmap

Phase 1: MVP (Months 1–2)

Goal: Voice-first loan audit + daily bookkeeping for Mann Deshi women via WhatsApp and IVR

- Agent orchestrator with raw function calling on Gemini
- 3 core skills: Loan Term Extractor (vision-based), NBFC Verifier, Rate Calculator
- Bookkeeping Assistant skill: voice-input income/expense logging with daily summaries
- Gemini end-to-end voice pipeline (STT → reasoning → TTS) for Marathi and Hindi
- WhatsApp Business API integration via BSP (Gupshup recommended for India presence)
- IVR setup via Exotel or Knowlarity with toll-free number
- Privacy consent flow for WhatsApp users (document processing disclosure + opt-in)
- DhanKanya RAG integration for basic financial literacy Q&A
- Manual testing with 50–100 real loan agreements from Mann Deshi's portfolio
- Pilot with 50–100 Mann Deshi SHG members

Phase 2: Enhanced Intelligence (Months 3–4)

Goal: Add Red Flag Scanner, expand languages, pilot SHG group features

- Red Flag Scanner skill: predatory clause pattern detection
- Telugu language support added to voice pipeline
- State-specific guidance templates for Maharashtra, Andhra Pradesh, Karnataka
- SHG Group Mode: multi-user bookkeeping under group leader's session
- Post-audit affordability check (EMI auto-populated into bookkeeping)
- Channel usage analytics: measure WhatsApp vs. IVR engagement to inform Phase 3 channel investment
- Beta expansion to 500–1,000 users across 20+ SHG groups

Phase 3: Scale and Compliance (Months 5–6)

Goal: Security audit, NGO partnerships, native app launch

- Native Android app with offline mode and visual risk dashboard
- Security audit and penetration testing across all channels
- WhatsApp auto-deletion compliance verification
- Partnership with 3 financial literacy NGOs
- Grievance Filing skill: draft RBI Ombudsman complaints
- Public launch targeting 10,000+ active users (WhatsApp + IVR + app)

Phase 4: Ecosystem Integration (Months 7–12)

Goal: Credit bureau integration, B2B pilots, national expansion

- Credit Bureau API integration (with user consent) for holistic credit health view
- B2B white-label pilot with 2 ethical NBFCs or cooperatives
- Government Scheme Matcher skill: surface relevant central/state schemes
- Loan Comparison Engine skill: alternative lender suggestions
- Impact measurement study with academic partner
- National expansion beyond Maharashtra via NABARD and microfinance network partnerships

14. Team Requirements

Role	Scope
Technical Architect (Pranay Tiwari)	Overall technical direction; agentic architecture design; LLM pipeline; skill design and implementation
Product / Domain Lead (Priyanka Kamath)	Mann Deshi relationship; India-based user research; SHG ecosystem expertise; PRD refinement; contributor recruitment
Program Lead (Nurtekin Savas)	Strategic oversight; partnership facilitation; resource coordination
ML/AI Engineer (1–2)	Gemini integration; voice pipeline (STT/TTS); RAG pipeline; skill function implementation; model evaluation
Backend Engineer (1)	Agent orchestrator; WhatsApp BSP integration; IVR telephony integration; ephemeral document processing; session management
Mobile Developer (1)	Native Android app (Phase 3); offline-first architecture; on-device model deployment
UX/Voice Designer (1)	Vernacular conversation design; IVR flow design; WhatsApp message templates; low-literacy interaction patterns
Regulatory Analyst (part-time)	RBI guidelines interpretation; NBFC database maintenance; monthly compliance updates
Growth/Community Lead (1)	SHG network distribution; NGO partnerships; Mann Deshi co-branding; user feedback loops

15. Appendix A: Key Regulations Reference

RBI Digital Lending Guidelines 2022

- All loan disbursements must flow directly to borrower’s bank account (no intermediary wallets)
- LSPs (Lending Service Providers) must be registered
- Borrowers entitled to cooling-off period for certain digital loans
- Interest rates must be disclosed as APR, not flat rates
- Recovery agents must not use coercive practices

NBFC Fair Practices Code

- Transparent disclosure of all charges in loan agreement
- No penalty for prepayment on certain floating-rate loans
- Grievance redressal mechanism mandatory

WhatsApp Business API Compliance

- SmartLend must register as a business (not AI provider) on WhatsApp Business Platform
- Template messages must be approved by Meta before proactive outreach
- User opt-in required before sending template messages
- 24-hour customer service window for free-form messages

16. Appendix B: DhanKanya Integration Reference

DhanKanya (github.com/100-GIGA/DhanKanya) is an open-source financial literacy chatbot developed by 100GIGA. SmartLend adopts its ChromaDB + BGE embeddings RAG pipeline, state_templates.json regional guidance framework, speech recognition integration, and expense tracking data model. Key modifications: knowledge base repopulated with RBI lending guidelines and NBFC data; Claude API replaced with model-agnostic LLM layer (Gemini recommended); Streamlit UI replaced with WhatsApp/IVR/native interfaces; expense tracker enhanced with post-audit EMI integration.